

Positive Definite Matrix

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Quadratic Function

$$f(x, y) = 9x^2 + 4xy + y^2 = \begin{bmatrix} x \\ y \end{bmatrix}^T \underbrace{\begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x \\ y \end{bmatrix}$$

Eigenvalue decomposition:

$$\mathbf{A} = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^T$$

with eigenvalues:

$$\lambda_1 = 9.47 \quad \lambda_2 = 0.53$$

and eigenvectors:

$$\mathbf{u}_1 = \begin{bmatrix} 0.97 \\ 0.23 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} -0.23 \\ 0.97 \end{bmatrix}$$

Quadratic Function

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```
1 import numpy as np
2
3 A = np.array([[9, 2], [2, 1]])
4 eig_val, eig_vec = np.linalg.eig(A)
5
6 print('lambda_1 = ', eig_val[0])
7 print('lambda_2 = ', eig_val[1])
8 print('u_1 = ', eig_vec[:, 0])
9 print('u_2 = ', eig_vec[:, 1])
```

Quadratic Function

$$f(x, y) = 9x^2 + 4xy + y^2 = \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}^\top}_{\mathbf{w}^\top} \underbrace{\begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{w}}$$

Using the eigenvalue decomposition:

$$\begin{aligned} \mathbf{w}^\top \mathbf{A} \mathbf{w} &= \mathbf{w}^\top (\lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top) \mathbf{w} \\ &= \lambda_1 (\mathbf{u}_1^\top \mathbf{w})^2 + \lambda_2 (\mathbf{u}_2^\top \mathbf{w})^2 \end{aligned}$$

with eigenvalues and eigenvectors:

$$\lambda_1 = 9.47 \quad \mathbf{u}_1 = \begin{bmatrix} 0.97 \\ 0.23 \end{bmatrix} \quad \lambda_2 = 0.53 \quad \mathbf{u}_2 = \begin{bmatrix} -0.23 \\ 0.97 \end{bmatrix}$$

Thanks for your attention!

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